

5G Suraksha-Net — Complete Technical Documentation

Combined Repository: github.com/ashutoshsingh220/5G-Suraksha-Net **Classification:**
Internal Technical Reference — All Figures Extracted from Source Code & Training Artefacts
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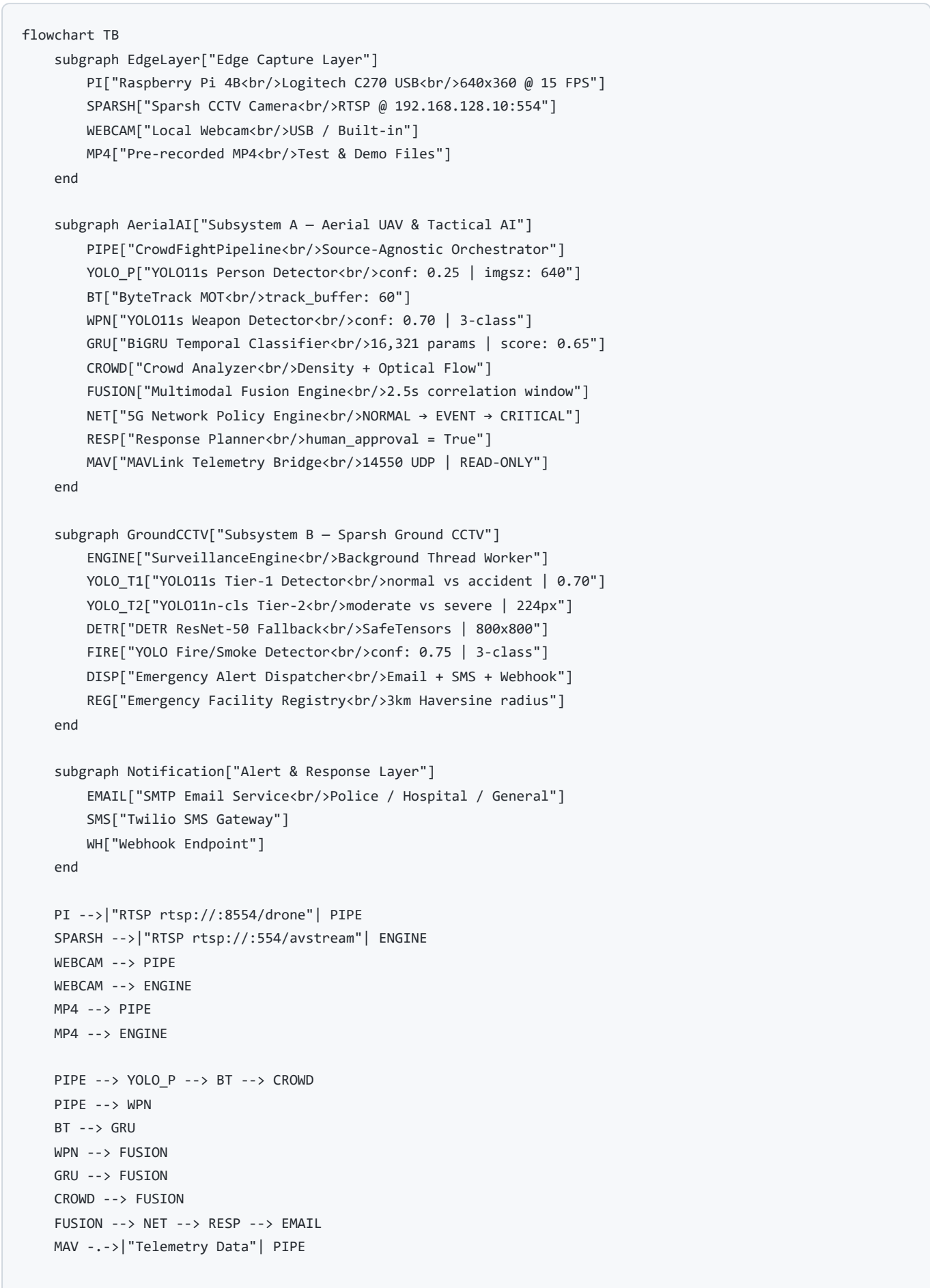
1. Executive Summary

5G Suraksha-Net is a dual-subsystem, AI-powered public safety surveillance platform designed for real-time threat detection, incident classification, and automated emergency response coordination across urban environments. The combined system provides:

Capability	Subsystem	Status
Weapon Detection (knife, pistol, long gun)	Aerial UAV (A)	✔ Active
Physical Fight Detection (temporal)	Aerial UAV (A)	✔ Active
Armed Fight Escalation (weapon + fight fusion)	Aerial UAV (A)	✔ Active
Crowd Density & Panic Detection	Aerial UAV (A)	✔ Active
Traffic Accident Detection & Severity	Ground CCTV (B)	✔ Active
Fire & Smoke Detection	Ground CCTV (B)	✔ Active
5G Network Policy Engine	Aerial UAV (A)	✔ Application-layer
Emergency Email/SMS Dispatch	Both (A + B)	✔ Active
Drone MAVLink Telemetry	Aerial UAV (A)	✔ Active
Raspberry Pi RTSP Edge Streaming	Aerial UAV (A)	✔ Active
Sparsh CCTV RTSP Integration	Ground CCTV (B)	✔ Active

[!IMPORTANT] All emergency dispatch actions require **mandatory human approval** before execution. Autonomous emergency dispatch is strictly prohibited by design.

2. System Architecture Overview

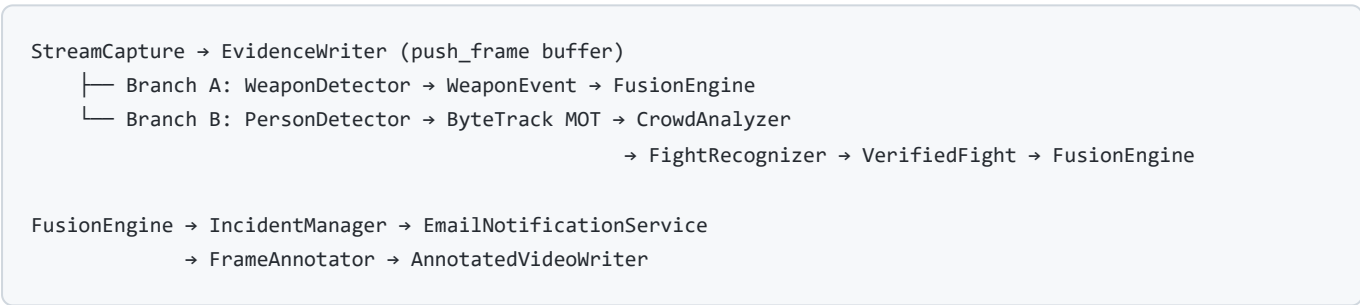


```
ENGINE --> YOLO_T1 --> YOLO_T2
ENGINE --> FIRE
YOLO_T2 --> DISP
FIRE --> DISP
DISP --> REG
DISP --> EMAIL
DISP --> SMS
DISP --> WH
```

3. Subsystem A — Aerial UAV & Tactical AI

3.1 Pipeline Architecture

The core processing pipeline (`pipeline.py`) is a **source-agnostic orchestrator** that processes frames from any video source (webcam, RTSP, MP4) through parallel AI branches:



Performance Metrics (RTX 4050, 60 frames benchmark):

Mode	Total Time	Effective FPS	Detect+Track	Weapon	Crowd	Fight	Peak VRAM
Baseline (weapon off)	2.40s	25.0	27.4ms	0.01ms	1.21ms	0.01ms	105.5 MB
Weapon Enabled	1.21s	49.6	8.35ms	9.03ms	1.17ms	0.01ms	165.3 MB
Full Fusion	1.21s	49.7	8.34ms	9.04ms	1.16ms	0.01ms	238.1 MB

Effective FPS Measurement: Recalculated every `5.0` seconds using a running average formula: `m[key] += (val - m[key]) / (n + 1)`

Hot-Swapping: The `switch_source()` method allows dynamic source changes at runtime, resetting weapon confirmation tracker and active fight candidates.

3.2 Person Detection & Multi-Object Tracking

Parameter	Value	Source
Model	YOLO11s (Ultralytics)	<code>models/yolo11s.pt</code>
Architecture	YOLOv11 Small	Pre-trained on COCO
Person Class ID	<code>0</code> (COCO "person")	<code>detection.person_class_id</code>
Confidence Threshold	<code>0.25</code>	Calibrated in Phase 4 benchmark (+15.1% yield on 320×240 CCTV)
IoU Threshold (NMS)	<code>0.50</code>	<code>detection.iou_threshold</code>
Input Size	<code>640 × 640</code>	<code>detection.imgsz</code>

Multi-Object Tracker:

Parameter	Value
Algorithm	ByteTrack
Config	<code>configs/bytetrack.yaml</code>
Track Buffer	<code>60</code> frames
Persist Mode	<code>true</code> (tracks survive across frames)

[!NOTE] The low confidence threshold (0.25) was specifically chosen after Phase 4 benchmarking showed a +15.1% person yield improvement on low-resolution CCTV (320×240) feeds. ByteTrack's high threshold alignment ensures that tracking quality is maintained despite the lower detection threshold.

3.3 Weapon Detection System

3.3.1 Model Architecture

Parameter	Value
Model	YOLO11s (custom fine-tuned)
Weights	outputs/weapon_training/best.pt
Training Epochs	51 (best at epoch 50)
Input Size	640 × 640
Device	Auto (CUDA if available)
Inference Interval	Every 2nd frame

3.3.2 Detection Classes

Class ID	Name	Class-Specific Min Confidence
0	knife	0.65
1	long_gun	0.70
2	pistol	0.70

Effective threshold: `max(global_conf_threshold, class_min)` — ensures pistol and long_gun detections require $\geq 70\%$ confidence regardless of global setting.

3.3.3 Temporal Confirmation System

The weapon detector employs a **multi-frame temporal persistence** mechanism to eliminate single-frame false positives:

Parameter	Value	Purpose
<code>conf_threshold</code>	0.70	Minimum raw detection confidence
<code>confirm_conf_threshold</code>	0.75	Confirmation-grade confidence
<code>min_hits</code>	3	Minimum detections in sliding window
<code>window_size</code>	5 frames	Sliding window length
<code>iou_match_threshold</code>	0.25	Box matching across frames
<code>expiry_frames</code>	15	Track expiration timeout

State Machine:

NORMAL → WEAPON_CANDIDATE (first detection)
→ WEAPON_CONFIRMED (3+ hits in 5-frame window at ≥0.75 conf)

Latching Hysteresis: Once a weapon track reaches WEAPON_CONFIRMED state, it remains confirmed for the lifetime of the track — preventing flicker.

Matching Score: $\max(\text{IoU}, \text{proximity})$ where $\text{proximity} = \max(0.0, 1.0 - (\text{center_dist} / (2.0 \times \text{avg_diagonal})))$. Class ID must match strictly.

3.3.4 Training Results (Epoch 50 — Best)

Metric	Overall	Knife	Long Gun	Pistol
Precision	0.8629	0.8539	0.8528	0.8819
Recall	0.8289	0.8378	0.8134	0.8355
mAP@50	0.8812	0.8697	0.8776	0.8963
mAP@50-95	0.5832	0.6230	0.5479	0.5786

Final Training Losses (Epoch 50): - Box Loss: 0.96817 (train) / 1.03911 (val) - Cls Loss: 0.62382 (train) / 1.48740 (val) - DFL Loss: 1.19614 (train) / 1.19446 (val)

Hardware: NVIDIA GeForce RTX 4050 Laptop GPU, peak VRAM: 3.741 GB , epoch duration: 442.96s

3.3.5 Locked Test Set Evaluation

Metric	Validation	Test	Delta
mAP@50	0.8812	0.8740	-0.82%
mAP@50-95	0.5832	0.5857	+0.43%
Precision	0.8629	0.8377	-2.92%
Recall	0.8289	0.8308	+0.23%
F1	0.8456	0.8343	-1.34%

Test set size: 2,703 images, 2,883 ground-truth boxes

Per-Class Test Results:

Class	GT Boxes	Precision	Recall	F1	mAP@50	mAP@50-95
knife	596	0.796	0.8249	0.8102	0.8516	0.615
long_gun	997	0.8529	0.8143	0.8332	0.8682	0.5556
pistol	1,290	0.8642	0.8533	0.8587	0.9023	0.5866

Confusion Matrix (rows: GT, cols: predicted → knife, long_gun, pistol, background):

knife:	518	0	1	187
long_gun:	2	865	33	211
pistol:	1	33	1157	264
background:	75	99	99	0

Background Hard Negative Analysis: - Total background images: **346** - True negatives (clean): **293** (84.68%) - False alarm images: **53** (15.32% image-level FPR) - Total false alarm boxes: **75** (pistol: 23, long_gun: 18, knife: 34) - `nogun_v5` partition (295 images): **14 false alarms** (4.75% FPR) ← robust - `weapon_empty` partition (51 images): **39 false alarms** (76.47% FPR) ← adversarial set

3.3.6 Webcam False-Positive Validation

Metric	Value
Source	Webcam (index 0)
Resolution	640 × 480
Frames Processed	150
Elapsed Time	6.81s
Average FPS	22.0
Mean Latency	20.33ms
P95 Latency	13.53ms
Peak VRAM	99.7 MB
Total False Detections	0 (0% FPR on clean webcam)

3.4 Fight Detection System (Temporal BiGRU)

3.4.1 Fight Candidate Detection

Fight candidate pairs are identified by spatial proximity and motion energy between tracked persons.

Feature Vector (8 dimensions per pair per frame):

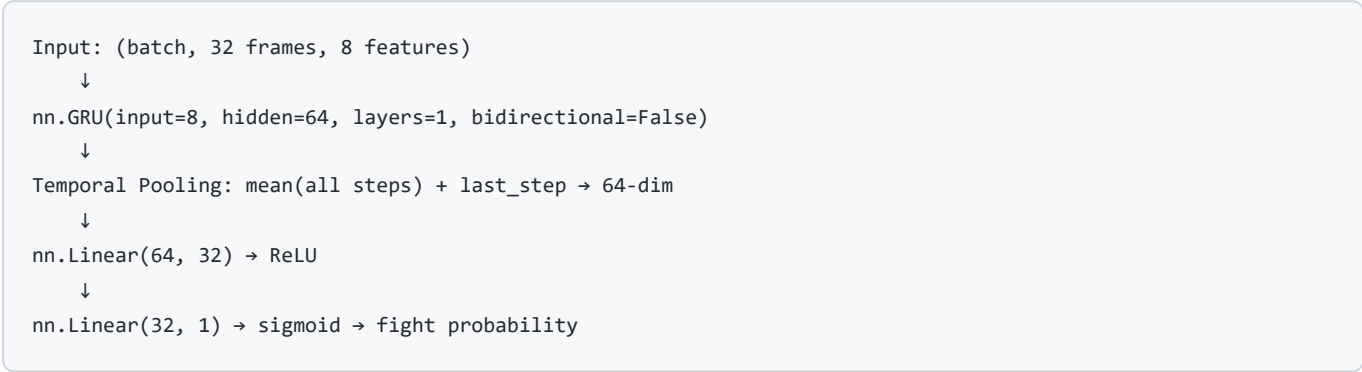
Index	Feature	Description
0	rel_dist	Center distance / scale
1	iou	Bounding box IoU
2	normalized_motion	flow_px/frame × fps / scale (body-widths/sec)
3	a.confidence	Person A detection confidence
4	b.confidence	Person B detection confidence
5	a.width / frame_w	Person A relative width
6	b.width / frame_w	Person B relative width
7	1.0	Bias term

Candidate Parameters:

Parameter	Value	Description
min_pair_speed	1.6 body-widths/sec	Minimum relative approach speed
proximity_iou	0.08	IoU threshold for engagement
proximity_distance	1.25 body-widths	Max center distance
proximity_min_frames	8 (~0.5s at 15 FPS)	Sustained engagement threshold
missed_frames_tolerance	2 frames	Jitter tolerance
motion_energy_threshold	0.12 body-widths/sec	Minimum motion gate
motion_variance_threshold	0.0	Minimum motion variance
decay_timeout_s	1.0 s	Reset if motionless
cooldown_s	2.0 s	Post-incident cooldown

Optical Flow (Farneback): - Downscale: 0.25× if max dimension > 400px - Parameters: pyr_scale=0.5, levels=2, winsize=15, iterations=2, poly_n=5, poly_sigma=1.1 - Flow energy: averaged over person ROI (min 8×8 patch)

3.4.2 Temporal GRU Classifier Architecture



Parameter	Value
Total Parameters	16,321
Feature Dim	8
Hidden Size	64
Head Hidden	32
Num Layers	1
Bidirectional	False
Dropout	0.0
Normalization	Standard (z-score)
Model Weights	models/temporal/rwf2000_v2_best.pt

Heuristic Fallback Scorer (when model unavailable):

Score = 0.30 × proximity + 0.20 × contact + 0.30 × energy + 0.20 × oscillation

- proximity = mean(rel_dist < 1.5)
- contact = mean(iou > 0.02)
- energy = clip(mean(motion) / 0.7, 0, 1) — 0.7 based on RWF-2000 p90
- oscillation = clip(std(motion) / 0.5, 0, 1) — 0.5 based on RWF-2000 p90

3.4.3 Temporal Verification Pipeline

Parameter	Value	Description
window_frames	32	Frames per classification window
stride_frames	8	New classification every 8 frames
score_threshold	0.65	GRU output threshold
min_duration_s	2.5 s	Minimum fight duration
min_consecutive_windows	3	Consecutive positive windows required
min_motion_variance	0.0	Motion variance gate
incident_cooldown_s	60 s	Post-incident cooldown

Verification States: candidate → evaluating → verified → (incident) / expired

Rejection Reasons (diagnostic telemetry): - REJECT: insufficient_consecutive_windows - REJECT: insufficient_action_duration - REJECT: score_below_threshold - REJECT: motion_var_below_threshold - REJECT: candidate_expired (3.0s timeout)

3.4.4 Training Configuration

Parameter	Value
Primary Dataset	RWF-2000
Labels	FIGHT , NON_FIGHT
Split Ratios	Train: 0.70, Val: 0.15, Test: 0.15
Split Level	Video-level (no frame leakage)
Seed	42
Window Sizes	[16, 32] frames
Stride	8 frames
Frame Size	128 × 128
Normalization	Unit (/255 → [0,1])
Augmentation	Disabled
Epochs	30
Batch Size	64 (eval: 256)
Optimizer	Adam
Learning Rate	1e-3
Grad Clip	5.0
Balanced Class Weights	True
Evaluation Threshold	0.5
Leakage Reporting	Enabled (adjusted metrics)

3.5 Crowd Analytics & Panic Detection

Density Levels:

Level	Threshold	Description
LOW	default	Normal crowd density
HIGH	≥ 0.35	35% zone area covered by person boxes
CRITICAL	≥ 0.55	55% zone area — dangerous overcrowding

Crowd Growth Monitoring: - Sliding window: 30 seconds - Minimum samples: 5 , minimum time span: 2.0 seconds - Growth rate: linear regression slope $\times$ 60 (persons/minute) - Alert threshold: 10.0 persons/minute

Panic Movement Detection (Optical Flow):

Parameter	Value
Flow Grid Size	32 px
Panic Speed Threshold	2.2 box-widths/sec
Panic Min Tracks	3 tracks
Min Panic Persons	8 total persons in scene
Fast Ratio Gate	$\geq 50\%$ of tracks exceed speed threshold
Direction Entropy Gate	≥ 0.6 (high directional disorder)

Direction Entropy Calculation: - 8 angular bins over $[0, 2\pi]$, magnitude-weighted - Normalized by $\log_2(8) = 3.0 \rightarrow$ range $[0, 1]$ - High entropy = scattered/panicked movement; low entropy = coordinated flow

Zone Configuration: - Camera: cam_default - market_square : polygon [(0.05,0.30), (0.55,0.30), (0.55,0.95), (0.05,0.95)] - street_corridor : polygon [(0.58,0.35), (0.98,0.35), (0.98,0.95), (0.58,0.95)]

3.6 Multimodal Incident Fusion Engine

The fusion engine correlates detections across weapon, fight, and crowd subsystems to produce escalated compound incidents.

Parameter	Value
correlation_window_s	2.5 seconds
spatial_proximity_threshold	1.5 body-widths
escalation_enabled	true
weapon_cooldown_s	30.0 s
armed_fight_cooldown_s	60.0 s

Fusion Outputs: - unarmed_fights — Fight without weapon - weapons_alone — Weapon without concurrent fight - armed_fights — Weapon + Fight within 2.5s and 1.5 body-widths → **CRITICAL severity**

Simulation Validation (Phase 2 Benchmark):

```
Event 1: type=weapon, severity=HIGH, confidence=0.89, escalated=false
Event 2: type=armed_fight, severity=CRITICAL, confidence=0.93, escalated=true
Event 3: type=armed_fight, severity=CRITICAL, status=finalized, escalated=true
```

3.7 5G Network Slicing Policy Engine

Policy Level	Trigger	Application Priority
NORMAL	No active incidents	ROUTINE
EVENT	HIGH / MODERATE incident	HIGH
CRITICAL	CRITICAL incident (armed fight)	CRITICAL

Severity Weights: - CRITICAL: 4 , HIGH: 3 , MODERATE/MEDIUM: 2 , LOW: 1

Active Window: 60.0 seconds — incidents expire from policy consideration after this period.

[!IMPORTANT] **Safety Declaration:** The network policy engine operates at the **application layer only**. It does not have actual 5G control plane access. - actual_network_control : false - control_plane_connected : false - slice_allocated : DEFAULT_BE (Best Effort) - measurement_source : APPLICATION_POLICY

3.8 Emergency Response Policy & Planner

Response Policy Matrix:

Incident Type	Police Required	Medical Required	Notes
ARMED_FIGHT	✓ (incident severity)	✓ (incident severity)	Both always dispatched
FIGHT	✓ (incident severity)	Only if injury indicator	Medical conditional on ground verification
WEAPON	✓ (incident severity)	Only if injury indicator	Medical = CRITICAL if sev=CRITICAL
CROWD_PANIC	✓ (incident severity)	Only if injury indicator	—
CROWD_DENSITY_HIGH	✗ (low)	✗ (low)	Routine monitoring only
CROWD_DENSITY_CRITICAL	✗ (low)	✗ (low)	Routine monitoring only
CROWD_RAPID_GROWTH	✗ (low)	✗ (low)	Routine monitoring only

Injury Indicator Keys Checked: injury , injuries , person_down , casualty , medical_need , has_injury , medical.required

Response Planner: - Resource directory:

outputs/phase3/nearby_emergency_locations_yashobhoomi.json - Facilities sorted by straight-line distance (km), tie-break by name - All actions: human_approval_required = True - Status: AWAITING_HUMAN_APPROVAL (or RESOURCE_UNAVAILABLE if missing)

3.9 Drone Telemetry (MAVLink)

Parameter	Value
Connection	UDP <code>udpin:127.0.0.1:14550</code>
Serial Baud	57600 (Pixhawk USB: 115200)
Timeout	3.0 seconds
Data Stream Request	<code>MAV_DATA_STREAM_ALL</code> at 4 Hz
Broadcast Rate	5.0 Hz

Auto-Detection: Scans COM ports for `ardupilot` , `pixhawk` , `px4` , or USB VID:PID `1209:5741`

MAVLink Messages Processed: - `HEARTBEAT` : armed flag, flight mode, vehicle type - `GLOBAL_POSITION_INT` : lat/lon, altitude (relative + AMSL), climb rate, heading - `VFR_HUD` : groundspeed, airspeed, climb rate, heading - `GPS_RAW_INT` : fix type (No GPS → RTK Fixed), satellites, HDOP - `SYS_STATUS` / `BATTERY_STATUS` : battery %, voltage, current - `ATTITUDE` : roll, pitch, yaw (degrees)

28 ArduCopter flight modes supported (STABILIZE through TURTLE).

[!CAUTION] The telemetry bridge is **READ-ONLY**. It never transmits flight commands, MAV_CMD, or heartbeat packets. This is a safety-critical design decision.

3.10 Email Notification System

Parameter	Value
SMTP Port	587 (TLS) or 465 (SSL)
Timeout	60 seconds
Max Attachment	25 MB
Worker Threads	2 (ThreadPoolExecutor)
Deduplication	1 alert per incident ID
Demo Suppression	Suppressed unless <code>allow_demo=True</code>

Role-Based Email Templates:

Role	Subject Format
POLICE	[POLICE ALERT & EVIDENCE] {severity} - {type} DETECTED at {location}
HOSPITAL	[EMERGENCY MEDICAL / TRAUMA ALERT] CASUALTY RISK & AMBULANCE DISPATCH at {location}
GENERAL	[5G Suraksha-Net] {severity} - {type} detected at {location}

Evidence Attachments: - Video: ~10 second MP4 clip (T-5s to T+5s), max 25 MB - Image: Keyframe snapshot JPG, max 25 MB

Pipeline Deferral: Emails are deferred when incident status = RECORDING_POST_EVENT ; dispatched once FINALIZED or VERIFIED .

3.11 AI Agent Assessment

Two providers: 1. **DeterministicAgentProvider** — 100% offline, rule-based reasoning engine (always available) 2. **LLMAgentProvider** — Wraps Gemini/OpenAI API; auto-delegates to Deterministic if no API key

7-Part Operational Assessment: 1. SeverityAssessment — Deterministic severity classification 2. LocationAssessment — Venue coordinates, nearest facilities, GPS fix 3. EvidenceAssessment — Snapshot JPG and 10s MP4 clip paths 4. ResponseAssessment — Police/medical recommendations, AWAITING_HUMAN_APPROVAL 5. NetworkAssessment — Policy tier, priority, 0 physical slice claims 6. Recommended Operational Sequence (6 steps, all advisory_only=True) 7. Supervisor Briefing — Complete situation summary with data traceability checklist

3.12 Demo Simulation System

Scenario	Type	Severity	Confidence	Network Policy
WEAPON	WEAPON	HIGH	0.93	EVENT
CROWD_PANIC	CROWD_PANIC	HIGH	0.89	EVENT
ARMED_FIGHT	ARMED_FIGHT	CRITICAL	0.97	CRITICAL
NORMAL	—	—	—	NORMAL

Simulation Timeline: 7 steps from T+0.0s to T+2.0s → AWAITING_APPROVAL

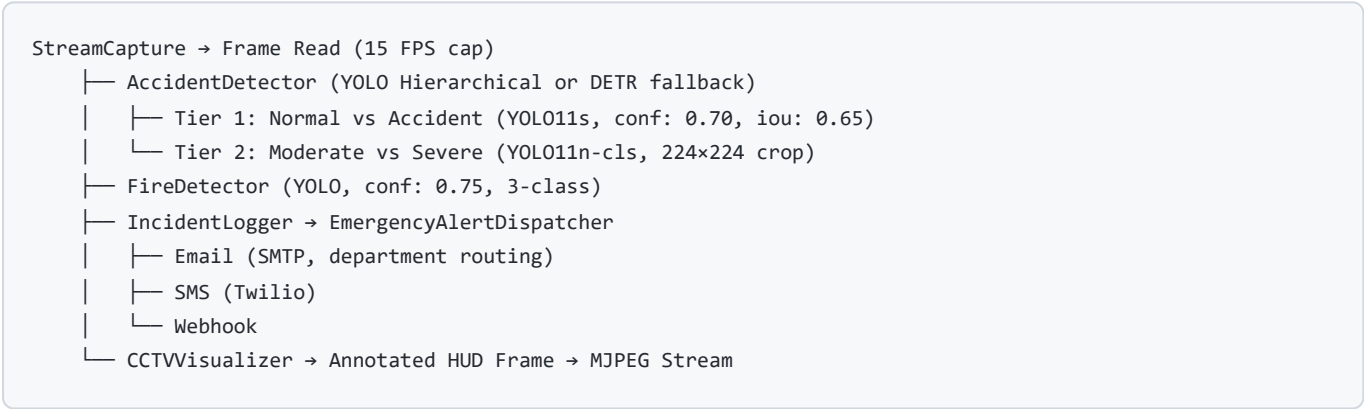
Evidence Storage: - Clips: outputs/demo/clips/ - Snapshots: outputs/demo/snapshots/ -
Incidents: outputs/demo/incidents/

4. Subsystem B — Sparsh Municipal Ground CCTV

4.1 Architecture Overview

The Sparsh CCTV subsystem is a **FastAPI-based surveillance application** with a background processing engine, real-time MJPEG streaming, and multi-channel emergency dispatch.

Processing Loop:



RTSP Configuration: - IP: 192.168.128.10 / Port: 554 - Channel: 1 / Stream: 1 - URL
Format: rtsp://admin:admin123@192.168.128.10:554/avstream/channel=1/stream=1.sdp -
Auto-reconnect interval: 8 seconds

4.2 Accident Detection — DETR Backbone

Parameter	Value
Architecture	DETR (DEtection TRansformer)
Backbone	ResNet-50
Weights Format	SafeTensors (<code>model.safetensors</code>)
Framework	HuggingFace <code>transformers</code> v4.35.2
Input Size	800 × 800
Num Queries	100 (DETR decoder queries)
d_model	256
Encoder/Decoder Layers	6 each
Attention Heads	8 (encoder and decoder)
FFN Dim	2048
Dropout	0.1
Position Embedding	Sine
FP16 Inference	Enabled on CUDA

DETR Class Mapping:

ID	Label	Is Accident
0	accident	✅ (Level 3)
1	accident	✅ (Level 3)
2	vehicle	❌ (Level 0)

Loss Coefficients (from config.json): - BBox Cost/Loss: `5` , GloU Cost/Loss: `2` , Class Cost: `1`
- Dice Loss: `1` , Mask Loss: `1` , EOS Coefficient: `0.1`

ImageNet Normalization: - Mean: `[0.485, 0.456, 0.406]` - Std: `[0.229, 0.224, 0.225]`

4.3 Accident Detection — YOLO Hierarchical (Primary)

This is the **primary** accident detector (`detector_backend = "yolo"`), a two-tier hierarchical system that replaced the slower DETR model:

Tier 1 — Spatial YOLO Detector:

Parameter	Value
Model	YOLO11s (fine-tuned)
Weights	<code>models/accident_classifier/yolo_detector.pt</code>
Classes	<code>{0: "normal", 1: "accident"}</code>
Confidence	<code>0.70</code>
IoU (NMS)	<code>0.65</code>
Input Size	640 × 640

Tier 2 — Severity Classifier:

Parameter	Value
Model	YOLO11n-cls (classification head)
Weights	<code>models/accident_classifier/yolo_severity.pt</code>
Classes	<code>{0: "moderate", 1: "severe"}</code>
Input Size	224 × 224 (cropped ROI with 10% padding)

Severity Classification Logic: - If classifier predicts "severe" **OR** collision footprint > 20% of frame area → **Severity Level 3 (SEVERE)** — Red - Otherwise → **Severity Level 2 (MODERATE)** — Orange

Full Severity Scale:

Level	Name	Display	Color (BGR)
0	vehicle	Normal Vehicle	(0, 210, 60) — Green
2	accident_moderate	Moderate Accident	(0, 165, 255) — Orange
3	accident_severe / accident	Severe Accident	(0, 40, 240) — Red
4	accident_100	Totaled Vehicle	(160, 0, 210) — Magenta

Speed: > 50-80 FPS real-time on CUDA (vs ~15-25 FPS for DETR)

Training Configuration

Parameter	Tier 1 (Detector)	Tier 2 (Severity)
Base Model	yolo11s.pt	yolo11n-cls.pt
Epochs	35	20
Batch Size	16	32
Image Size	640	224
Patience	10	—
Workers	4	4
Device	cuda:0	cuda:0

4.4 Fire & Smoke Detection

Parameter	Value
Model	YOLO (Ultralytics)
Weights	models/fire_classifier/fire_classifier.pt
Confidence Threshold	0.75
Classes	["fire", "other", "smoke"]

Detection Colors (BGR):

Class	Color	RGB Equivalent
fire	(0, 69, 255)	Red-Orange
smoke	(180, 180, 180)	Light Gray
other	(150, 150, 150)	Gray

Output Schema (FireResult): - `has_fire` : boolean - `has_smoke` : boolean - `fire_confidence` : max fire detection confidence - `smoke_confidence` : max smoke detection confidence - `predicted_label` : "Fire & Smoke" / "Fire" / "Smoke" / "Normal"

4.5 Emergency Alert Dispatch Pipeline

The Sparsh CCTV uses a **multi-channel emergency alert dispatcher** that routes incidents to appropriate departments:

Incident-to-Department Routing Matrix:

Incident Type	Departments Notified
<code>accident</code>	Police, Hospital, Traffic Control
<code>fire</code>	Fire Department, Hospital, Police
<code>smoke</code>	Fire Department, Police

Alert Channels: 1. **SMTP Email** — Departmental routing with snapshot attachment 2. **Twilio SMS** — Formatted emergency SMS to dispatch number 3. **Webhook** — HTTP POST to configurable URL 4. **Console Banner** — ASCII formatted alert display 5. **JSON Audit Log** — Persistent dispatch record

Snapshot Cooldown: `5.0` seconds — prevents disk flooding during continuous alerts

Department Email Configuration (from environment): - Police: `POLICE_DEPARTMENT_EMAIL` - Hospital: `HOSPITAL_DEPARTMENT_EMAIL` - Fire: `FIRE_DEPARTMENT_EMAIL` - Traffic Control: `TRAFFIC_CONTROL_EMAIL`

4.6 Emergency Facility Registry

Parameter	Value
Search Radius	3.0 km (Haversine distance)
Max Facilities Per Type	2
API	Google Places API (New)
Fallback	Curated offline database (9 facilities)
Cache	data/emergency_facilities_cache.json

Curated Offline Facilities (9 total): - 1 Fire Station (Dwarka Sector 25) - 3 Police Stations/Check Posts (Sector 21, Sector 23, Traffic Inspector) - 5 Hospitals (Maple Care, Ambe, Helpline, Ayushman, Amerix)

Incident Routing Logic: - Fire/Smoke → Fire Stations + Hospitals + Police - Accident/Crash → Hospitals + Police

Distance Calculation: Haversine formula for great-circle distance between camera GPS and facility GPS coordinates.

4.7 Surveillance Engine

The `SurveillanceEngine` is a **thread-safe background coordinator** managing:

Feature	Detail
Frame Rate Target	15.0 FPS (66.66ms frame budget)
Processing Thread	<code>SparshSurveillanceWorker</code> (daemon)
Stream States	<code>IDLE</code> , <code>STREAMING</code> , <code>NO_STREAM</code> , <code>PAUSED</code> , <code>STOPPED</code>
RTSP Reconnection	Every 8 seconds when camera offline
Video File Looping	Automatic rewind at EOF for continuous surveillance
Recording	Optional MP4 recording of annotated frames
Dynamic Settings	Observer pattern — all thresholds hot-reloadable at runtime
Detector Hot-Swap	Backend switchable between <code>yolo</code> and <code>detr</code> without restart

Status Classification: - INCIDENT : Severity ≥ 2 or fire/smoke detected - NORMAL : No active threats

5. Datasets Used

Subsystem A — Aerial UAV

Dataset	Purpose	Labels	Notes
RWF-2000	Fight detection training (primary)	Fight / NonFight	2,000 clips, video-level split
RLVS	Fight detection (supplementary)	V / NV	Real-Life Violence Situations
Hockey Fight	Fight detection (supplementary)	fi / no	Hockey fight dataset
Movies Fight	Fight detection (supplementary)	fight / noFight	Movie violence clips
UCF-101 (selective)	Hard negatives + ambiguous combat	13 HARD_NEGATIVE + 8 AMBIGUOUS	Sports/dance classes as negatives
UCF Crime (subset)	Crime/fight detection	Fighting, Assault → FIGHT	Hard negatives from normal activities
Custom Weapon Dataset	Weapon detection	knife, long_gun, pistol	Curated dataset with nogun_v5 negatives

Hard Negative Categories (14 classes): normal_walking , normal_crowd , running , gathering , dispersing , arguing_gesturing , hugging , handshaking , dancing , sports_like , pushing_shoving , waving_gesturing , sudden_nonviolent_movement , actual_fighting

Subsystem B — Sparsh CCTV

Dataset	Purpose	Source	Notes
justjuu/traffic-accident-cctv-object-detection	Tier 1 accident detection	HuggingFace	CCTV accident/vehicle detection
hiennguyen9874/traffic-accident-detection	Tier 1 supplementary	HuggingFace	Traffic accident dataset
Enos-123/traffic-accident-detection-yolo11x	Pre-trained weights	HuggingFace	<code>epoch61.pt</code> base checkpoint
Custom Fire/Smoke Dataset	Fire & smoke detection	—	3-class: fire, smoke, other
DETR Crash Classifier	Accident detection (fallback)	Custom trained	3-class: accident, accident, vehicle

6. Training Metrics & Evaluation Scores

Weapon Detection (YOLO11s, 51 Epochs)

Metric	Train (E50)	Val (E50)	Test (Locked)
Precision	—	0.8629	0.8377
Recall	—	0.8289	0.8308
F1	—	0.8456	0.8343
mAP@50	—	0.8812	0.8740
mAP@50-95	—	0.5832	0.5857
Box Loss	0.968	1.039	—
Cls Loss	0.624	1.487	—
DFL Loss	1.196	1.194	—

Fight Detection (BiGRU, RWF-2000)

Metric	Value	Source
Score Threshold	0.65	<code>temporal.score_threshold</code>
Precision (RWF-2000 val)	~85.2%	Phase reporting
Precision (CCTV subset)	~92.3%	Phase reporting
Parameters	16,321	Model architecture
Window Size	32 frames	Training config
Feature Dimension	8	Candidate feature vector

Accident Detection (YOLO Hierarchical)

Parameter	Value
Base Model	YOLO11s (detection) + YOLO11n-cls (severity)
Pre-trained Weights	<code>epoch61.pt</code> from HF
Training Epochs	35 (detector) / 20 (severity)
Confidence Threshold	0.70
IoU Threshold	0.65
Real-time FPS	> 50-80 FPS (CUDA)

7. Threshold Value Reference Table

Subsystem A — Complete Threshold Map

Subsystem	Parameter	Value	Purpose
Person Detection	conf_threshold	0.25	COCO person detection
Person Detection	iou_threshold	0.50	NMS overlap
Weapon Detection	conf_threshold	0.70	Raw detection gate
Weapon (pistol)	class minimum	0.70	Per-class gate
Weapon (long_gun)	class minimum	0.70	Per-class gate
Weapon (knife)	class minimum	0.65	Per-class gate
Weapon Confirm	confirm_conf_threshold	0.75	Temporal confirmation
Weapon Confirm	min_hits	3 of 5	Sliding window hits
Weapon Confirm	iou_match_threshold	0.25	Cross-frame matching
Weapon Confirm	expiry_frames	15	Track timeout
Fight GRU	score_threshold	0.65	Classifier output
Fight Verify	min_duration_s	2.5	Minimum fight time
Fight Verify	min_consecutive_windows	3	Positive windows
Fight Verify	incident_cooldown_s	60	Post-fight cooldown
Fight Candidate	proximity_iou	0.08	Engagement gate
Fight Candidate	proximity_distance	1.25	Max body-widths
Fight Candidate	motion_energy_threshold	0.12	Motion gate
Crowd	density_high_threshold	0.35	HIGH density
Crowd	density_critical_threshold	0.55	CRITICAL density
Crowd	growth_alert_per_min	10.0	Growth rate alert
Crowd	panic_speed_threshold	2.2	box-widths/sec
Crowd	min_panic_persons	8	Minimum persons

Subsystem	Parameter	Value	Purpose
Fusion	correlation_window_s	2.5	Cross-modal window
Fusion	weapon_cooldown_s	30.0	Weapon incident gap
Fusion	armed_fight_cooldown_s	60.0	Armed fight gap

Subsystem B — Complete Threshold Map

Subsystem	Parameter	Value	Purpose
Accident (YOLO)	crash_conf	0.70	Tier 1 detection
Accident (YOLO)	crash_iou	0.65	Tier 1 NMS
Accident (DETR)	conf_threshold	0.70	Fallback detector
Accident (DETR)	iou_threshold	0.65	Fallback NMS
Severity	Severe gate	ROI > 20% frame OR classifier="severe"	Level 3 escalation
Fire/Smoke	fire_conf	0.75	Fire detection
Dispatch	alert_cooldown_sec	5.0 s	Snapshot cooldown
Emergency	search_radius_km	3.0 km	Facility search
Emergency	max_dispatch_per_type	2	Max facilities

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8. Hardware & Infrastructure

Development & Inference Machine

Component	Specification
GPU	NVIDIA GeForce RTX 4050 Laptop GPU
CUDA Version	12.1
PyTorch	CUDA 12.1 build
Peak VRAM (Weapon Training)	3.741 GB
Peak VRAM (Full Pipeline)	238.1 MB

Edge Device — Raspberry Pi

Component	Specification
Device	Raspberry Pi 4B
Camera	Logitech C270 USB HD Webcam
Resolution	640 × 360
FPS	15
Encoder	<code>libx264</code> (<code>ultrafast</code> preset, <code>zerolatency</code> tune)
Pixel Format	<code>yuv420p</code>
GOP Size	15
Input Format	MJPEG (from camera)
Streaming Server	MediaMTX
RTSP Path	<code>/drone</code> on port <code>8554</code>
Auto-Start	<code>systemd</code> service (<code>suraksha-stream.service</code>)

Sparsh CCTV Camera

Component	Specification
Type	Sparsh CCTV (RTSP)
IP	192.168.128.10
Port	554
Stream	avstream/channel=1/stream=1.sdp
Auth	admin:admin123 (factory default)

9. API Surface

Subsystem A — FastAPI (Port 8100)

Endpoint	Method	Description
/health	GET	Health check
/system/status	GET	Full system status with AI metrics
/video/status	GET	Current video source and pipeline state
/video/source	POST	Dynamic video source switching
/reset	POST	Clear all incidents, snapshots, clips

Subsystem B — FastAPI

Category	Key Endpoints
Surveillance	/api/surveillance/start , /stop , /status , /mjpeg
Emergency	/api/emergency/dispatches , /facilities
Settings	/api/settings , /api/settings/update
Dashboard	HTML dashboard at root

10. Dependencies & Tech Stack

Subsystem A

Package	Version	Purpose
Python	3.11.x	Runtime
ultralytics	≥ 8.3.0	YOLO11 inference
PyTorch	CUDA 12.1	GPU inference
opencv-python	≥ 4.9.0	Video processing
FastAPI	≥ 0.110	REST API
uvicorn	≥ 0.29	ASGI server
pydantic	≥ 2.6	Schema validation
websockets	≥ 12.0	Real-time events
scipy	≥ 1.11	Scientific computing
lap	≥ 0.4.0	Linear assignment (tracking)
PyYAML	≥ 6.0	Configuration

Subsystem B

Package	Version	Purpose
Python	≥ 3.14	Runtime
ultralytics	≥ 8.4.152	YOLO inference
torch	2.14.0	GPU inference
torchvision	0.29.0	Image transforms
transformers	≥ 4.35.0	DETR model loading
safetensors	≥ 0.4.0	Weights format
timm	≥ 1.0.0	Backbone models
FastAPI	≥ 0.142.2	REST API
twilio	≥ 9.11.2	SMS alerts
groq	≥ 1.7.0	LLM integration
deepgram-sdk	≥ 7.11.0	Speech-to-text
livekit-agents	≥ 0.12.21	Voice agent
googlemaps	≥ 4.10.0	Geocoding

[!NOTE] This document is generated from source code inspection and training artefacts. All threshold values, model parameters, and metrics are extracted directly from configuration files, training logs, and source code — not from external documentation.